MIL-C-70544A (AR)
28 January 1988
SUPERSEDING
MIL-C-70544 (AR)
5 September 1986

MILITARY SPECIFICATION

CARTRIDGE, 81MM, ILLUMINATING, M853A1, LOADING, ASSEMBLING, AND PACKING

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This Specification covers requirements and quality assurance provisions for the loading, assembling, and packing for one type of mortar cartridge designated as Cartridge, 81MM, Illuminating, M853A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-286 - Propellants, Solid, Sampling, Examination and Testing
MIL-STD-414 - Sampling Procedures and Tables For Inspection by Variables for Percent Defective
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing

2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this Specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.3)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9352621 - Cartridge, 81MM, Illuminating, M853A1 W/Fuze, MTSQ M772
9354333 - Container, Ammunition, Fiber, For Cartridge, 81MM, M819/M853A1
9354616 - Box, Packing for Container, Ammunition Fiber for Cartridge, 81MM, M819/M853A1
9362798 - Packing and Marking for Box, Ammunition, Packing for Cartridge, 81MM, M819 and M853A1
9362799 - Packing and Marking for Container, Ammunition Fiber for Cartridge, 81MM, M819 and M853A1

INSPECTION EQUIPMENT DRAWINGS

9201136 - Tunnel, Light
9201268 - Procedure, Light Output Measurement
9201390 - Procedure, Photocell Checkout
9247071 - Photocell
9200964 - Photometer Head With Cell
7257965 - Limit Check Gage

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings, specifications, standards and this specification.

3.2 Cartridge. The cartridge shall comply with all requirements on drawing (dwg.) 9352621 and associated dwgs. and with all requirements specified in applicable specifications, standards, and this specification.

3.3 Moisture.

3.3.1 First fire composition. The moisture content of the first fire shall not exceed 0.30 percent, when tested as specified in 4.4.5.3.1 or 4.4.5.3.3.

3.3.2 Illuminant composition. The moisture content of the illuminant composition shall not exceed 0.30 percent, when tested as specified in 4.4.5.3.2 or 4.4.5.3.3.

3.4 Static functioning of the illuminant assembly. The static burn time shall be not less than 60 seconds. The average candlepower for the best 50 seconds shall be not less than 550,000. Determination shall be as specified in 4.4.5.4 (See 6.8).

3.5 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.4.5.6.

3.5.1 Misfire. The cartridge shall not fail to fire from the weapon.

3.5.2 Fuze/separation. The fuze shall function and the tail cone shall separate from the body tube assembly.

3.5.3 Functioning. The cartridge shall function satisfactorily (see 6.5) and shall provide effective illumination (see 6.6) for 50 seconds, min.

3.5.4 Security of parts. There shall be no metal parts failure in the bore of the weapon or in flight.

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3.5.5. Horizontal range. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.5.6 Fuze function time. The fuze assembly shall function and shall comply with the fuze function time in Table I.

TABLE I. Fuze Function Time

Fuze setting (seconds)	Charge	Fuze function time (seconds)
13.5	1	13.5 ± 1.0
33.5	4	33.5 ± 1.0

3.5.7 Premature burst. The fuze shall not function in the bore of the weapon or in the first 100 meters of flight.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship.

3.7.1 Parts and assemblies. All parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion, and other foreign material that may adversely affect fit or function. The cleaning method used shall not be injurious to the parts nor shall any parts be contaminated.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Cloth. All cloth shall be free of tears, rips, and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears, or foreign matter.

3.7.4 Painting and marking. The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification (see 6.10).

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items in the quantities indicated in Table II.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, 81MM: Illuminating M853A1 Assembly	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	First Fire Composition (dwg. 9381079)	4 oz	100%	3.3.1	4.4.3.1/4.4.5.3.1/ 4.4.5.3.3
	Illuminant Composition (dwg. 9381079)	4 oz	100%	3.3.2	4.4.3.1/4.4.5.3.2/ 4.4.5.3.3
	Static Functioning (dwg. 9381079)	10	100%	3.4	4.4.3.2/4.4.5.4
	Projectile Loading Assembly (dwg. 9381093)				
	Examination for defects	50*	100%	3.2	4.4.2.4/4.4.2.5
	Cartridge (dwg. 9352621)				
	Examination for defects	50*	100%	3.2	4.4.2.6/4.4.2.7
	Ballistic Test (Proving Ground)	50*	100%	3.5	4.4.3.3/4.4.5.6

NOTES:

* Same group of samples.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot of Cartridges, 81MM, Illuminating shall contain:

a. Projectile body metal parts in accordance with the same drawing, same drawing revision, same specification, and same specification revision, of one lot interfix number, from one manufacturer.

b. Ignition cartridges, of one lot interfix number, from one manufacturer.

c. Fin assemblies, of one lot interfix number, from one manufacturer.

d. Propelling charges, made from not more than one lot of propellant, from one manufacturer.

e. Loaded fuzes of one interfix from one manufacturer.

f. Parachute assemblies from one lot interfix number from one manufacturer.

g. Canister assemblies from one manufacturer, in accordance with the same drawing and same drawing revision.

h. Obturating rings, of one lot interfix, from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9381084
4.4.2.1	Canister Illuminant Metal Parts Assembly				NEXT HIGHER ASSEMBLY 9381079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Point stake inadequate		0.65%	3.2	Visual/Manual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Canister Loading Assembly (Prior to Loading Illuminant Composition)	1 OF 1	9381079
			NEXT HIGHER ASSEMBLY 9381093

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Coating on inside surfaces of canister missing, inadequate or tacky		0.40%	3.2	Visual/Manual
Minor 201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Canister Loading Assembly (After Loading Illuminant Composition)	1 OF 1	9381079
			NEXT HIGHER ASSEMBLY 9331093

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ignition disc assembly missing		0.40%	3.2	Visual
102	Filler rings not flush to 1/16" above flush		0.40%	3.2	Gage
103	Swivel fails to rotate freely		100%	3.2	Visual/Manual
104	Static functioning test			3.4	4.4.3.2/4.4.5.4
105	First fire composition moisture content test			3.3.1	4.4.3.1.1/4.4.5.3.1
106	Illuminant composition moisture content test			3.3.2	4.4.3.1.2/4.4.5.3.2
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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AMSMC Form 1570, 1 Feb 85

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile, 81MM: Illuminating M853A1 Loading Assembly During Assembly Operation	1 OF 1	9381093
			NEXT HIGHER ASSEMBLY 9352621

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Shroud lines pulled out of parachute bag beyond max. drawing requirement		0.40%	3.2	Gage
102	Parachute bag closure cord not removed		0.40%	3.2	Visual
103	Pull cord improperly attached to parachute disc, parachute assembly or drogue plate		0.40%	3.2	Visual
104	Pull test for oval sleeve to suspension wire		Note 1	3.2	4.4.5.10
105	Parachute shroud loops not all attached to suspension wire loop		0.40%	3.2	Visual
Minor					
201	Canister loading assembly improperly seated		0.65%	3.2	Visual
202	Spacer felt misaligned or missing		0.65%	3.2	Visual
203	Cable loop attachment not above separator disc		0.65%	3.2	Visual
204	Separator disc improperly positioned or missing		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

Note 1: See paragraph 4.4.3.5

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Projectile, 81MM: Illuminating M853A1 Loading Assembly Including Tail Cone Assembly	1 OF 1	9381093
			NEXT HIGHER ASSEMBLY 9352621

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Warning missing, incomplete, or incorrect		100%	3.2	Visual
b	Obturator ring missing, Note 1		100%	3.2	Visual
Major					
101	Depth of rear shear pin holes, Note 2		0.40%	3.2	Gage
102	Location of rear shear pin holes from end of body tube		0.40%	3.2	Gage
103	Diameter of rear shear pin holes, Note 2		0.40%	3.2	Gage
104	Shear pin(s), rear missing or loose		100%	3.2	Visual
105	Shear pin(s) not flush or below		100%	3.2	Visual
106	Protective dust plug missing (if applicable)		0.40%	3.2	Visual
107	Obturator ring not flush or below, Note 1		100%	3.2	Gage
108	Tack weld missing, Note 1		100%	3.2	Visual
109	Tail cone not fully seated		0.40%	3.2	Gage
110	Pull cord pinched between drogue plate and split sleeves		0.40%	3.2	Visual
111	Excess silastic present on OD of body tube or tail cone		0.40%	3.2	Visual
Minor					
201	Markings incorrect, missing or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: 1. If attachment of the obturator ring is done during cartridge assembly, these defects will be inspected at that time.
2. Alternate sampling plan may be used as specified in paragraph 4.4.5.9.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cartridge, 81MM Illuminating M853A1 W/Fuze MTSQ, M772. Prior to Assembly of Fuze Fin Assembly & Ignition Cartridge	1 OF 1	9352621
			NEXT HIGHER ASSEMBLY 9354333

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Damaged threads on fuze		0.40%	3.2	Visual
102	Damaged threads on ignition cartridge		0.40%	3.2	Visual
103	Damaged threads on cartridge housing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Cartridge, 81MM Illuminating 853A1 W/Fuze MTSQ, M772	1 OF 2	9352621
			NEXT HIGHER ASSEMBLY 9362799

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Ignition cartridge missing, cracked or damaged		100%	3.2	Visual
b	Propelling charge increment(s) missing, cracked or damaged		100%	3.2	Visual
c	Fuze safety wire missing		100%	3.2	Visual
d	Fuze not set on safe		100%	3.2	Visual
Major					
101	Assembly fails to freely traverse alignment gage for 20 inches, min.		100%	3.2	Gage
102	Inadequate torque applied to fin boom		0.40%	3.2	Torque test/4.4.5.5
103	Inadequate torque applied to fuze		0.40%	3.2	Torque test/4.4.5.5
104	Damaged fin blades		0.40%	3.2	Visual/Magnifier
105	Security of parts		0.40%	3.2	Manual/4.4.5.7
106	Total length, max.		0.40%	3.2	Gage
107	Inadequate torque applied to ignition cartridge		0.40%	3.2	Torque test/4.4.5.5
108	Identifying strip missing		100%	3.2	Visual
Minor					
201	Protective finish damaged		0.65%	3.2	Visual
202	Markings incorrect, illegible or missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Cartridge, 81MM Illuminating M853A1 W/Fuze MTSQ, M772	2 OF 2	9352621
			NEXT HIGHER ASSEMBLY: 9362799

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	PROVING GROUND TEST				
	Ballistic Acceptance	50		3.5	4.4.3.3/4.4.5.6

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Container, Ammunition, Fiber, for Cartridges M819/M853A1/ Before Securing	1 OF 1	9362799
			NEXT HIGHER ASSEMBLY: 9362798

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parts not secure to cap or container		0.40%	3.2	Manual/4.4.5.7
102	Container damaged		0.40%	3.2	Visual
103	Absence of gap between cap and tube prior to taping		0.40%	3.2	Visual/5.1
Minor					
201	Marking incorrect, illegible or missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Container, Ammunition, Fiber, for Cartridges M819/M853A1/ After Securing	1 of 1	9362799
			NEXT HIGHER ASSEMBLY 9362798

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cartridge loose inside container		0.40%	3.2	Manual
102	Inner sealing strip not applied smoothly or oriented improperly		0.40%	3.2	Visual
103	Outer sealing strip not applied smoothly or oriented improperly		0.40%	3.2	Visual
104	Outer sealing strip not folded into tab at end		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.10	Box, Packing, Ammo for 81MM Mortar Prior to Closing	1	1	9362798

NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Containers not firmly packed in box		0.40%	3.2	Manual
Minor					
201	Box or nails loose		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70544A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Box, Packing, Ammo for 81MM Mortar After Closing	1 OF 1	9362798
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents of box loose		0.40%	3.2	Manual
102	Box damaged exposing contents		0.40%	3.2	Visual
103	Hardware or strapping missing or loose		0.40%	3.2	Visual
Minor					
201	Marking incorrect, illegible or missing		0.65%	3.2	Visual
202	Metallic seal improperly installed		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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4.4.3 Testing.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.3.1 Moisture content.

4.4.3.1.1 First fire composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce first fire sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.1.2 Illuminant composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce illuminant sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.2 Static functioning. The test shall be performed as specified in 4.4.5.4. A sample for the First Article inspection is specified in Table II. A sample for the inspection lot shall be selected in accordance with the sampling plan shown in 4.4.3.2.1. The candle loading assemblies shall be tested for static burning time and candlepower. If the static burning time is less than 60 seconds or the average candlepower for best 50 seconds is less than 550,000 candlepower, the inspection lots represented by the samples shall be rejected. No individual test sample shall have the candlepower less than 500,000 minimum in the best 50 seconds otherwise the inspection lot shall be rejected.

4.4.3.2.1 Sampling plan. An illuminant inspection lot shall be defined in paragraph 6.7.

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4.4.3.2.1.1 Sampling. A sample of three candles from each inspection lot (approximately 85 candles) shall be selected. The candle assemblies shall be tested for compliance with the requirements specified in 4.4.3.2. If the sample of three candles fails to comply the requirements specified in 4.4.3.2, a second sample of three candles shall be tested. If the second sample of three candles also fails to comply the requirements specified in 4.4.3.2, the inspection lot represented by the sample shall be rejected. This sampling frequency shall be continued until 15 consecutive lots (15 groups of 3 candles) comply with the specified requirements.

4.4.3.2.1.2 Sampling. After successful completion of 4.4.3.2.1.1, the sampling frequency may be reduced to three candles from each group of six inspection lots (approximately 510 candles). The canister assemblies shall be tested for compliance with the requirements in 4.4.3.2. If the sample of three candles from each group of six inspection lots fails to comply the requirement specified in 4.4.3.2, a second sample of three candles shall be tested. If the second sample of three candles also fails to comply the requirement specified in 4.4.3.2, the group of six inspection lots shall be rejected. This sampling frequency shall be continued until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.2.1.1.

4.4.3.2.1.3 Sampling. After successful completion of 4.4.3.2.1.2, the sampling frequency may be reduced to three candles from each group of fifteen inspection lots (approximately 1275 candles). The canister assemblies shall be observed for compliance with the requirements specified in 4.4.3.2. If the sample of three candles from each group of fifteen inspection lots fails to comply the requirements specified in 4.4.3.2, a second sample of three candles shall be tested. If the second sample of three candles also fails to comply the requirements specified in 4.4.3.2, the group of fifteen inspection lots shall be rejected. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.2.1.2.

4.4.3.2.1.4 Sampling. In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.4.3.2.1.2 and in the event of a production stoppage greater than two weeks, the sampling plan shall revert to 4.4.3.2.1.1. If the mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall be reverted to 4.4.3.2.1.2.

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4.4.3.2.1.5 Sampling. In the event of failure of any sample to comply with requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot.

4.4.3.3 Proving ground. Fifty sample cartridges shall be selected for this test. The test shall be performed as specified in 4.4.5.6. Defect classification and individual accept/reject criteria shall be in accordance with the following table:

Categories	Defect	Accept/Reject	
Critical:	Premature burst or metal parts separation	0	1
Major:	Misfire	1	2
Major:	Fuze non-function or non-separation of tail cone from body tube assembly	1	2
Major:	Horizontal range less than 80% of average range	0	1
Major:	Fuze functioning time (See Note 1) See para 4.4.3.3.1.1		
Major:	Illuminant fails to ignite or provide effective illumination for 50 seconds or less (See Note 2)	2	3

Note 1: If the fuze performance of any round in the sample exceeds the timing requirements specified, and the round functions properly in all respects with the exception that the "effective illumination" time is less than fifty seconds because the illuminant and parachute assembly hit the ground while still burning, the illumination time defects shall not be additionally assessed against that round.

Note 2: If a cartridge functions properly in all respects with the exception that the "effective illumination" time is less than fifty seconds because the illuminant and parachute assembly hit the ground while still burning and if the burst height, that is, the height above ground level at which the tail cone separated from the tube body was less than 475 meters the performance of the cartridge will not be considered in evaluating the acceptability of the lot for illumination time defects.

4.4.3.3.1 Overall acceptance criteria. In addition to the individual accept/reject criteria of paragraph 4.4.3.3, the lot shall be rejected if a total of four or more Major defects occur.

4.4.3.3.1.1 Acceptance criteria for fuze. The lot shall also be rejected if the percent lot defective for fuze functioning time as computed with the use of MIL-STD-414, Section B, with an AQL of 1.5 percent, exceeds the maximum allowable percent lot defective for each setting.

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4.4.3.4 Strength of heat weld. One obturating ring sample, minimum, after every 100 projectiles, shall be assembled to a fixture, introduced to the production line, heat sealed and tested as specified in 4.4.5.8. Failure of the sample to meet the requirement shall be cause for those 100 preceding projectiles represented by the sample to be rebanded and retested.

4.4.3.5 Pull test for oval sleeve crimps to suspension wire. (See drawing 8865018, 9381075 & 9381093) Pull test shall be conducted in accordance with paragraph 4.4.3.5.1 and 4.4.3.5.2.

4.4.3.5.1 Sampling testing plan. One assembly shall be selected at the start of each 4 hours of each production from each crimping machine to qualify the machine. A sample shall also be selected immediately following any adjustment to the machine which effects the crimping load regardless of when the adjustment is made. The test shall be performed as specified in 4.4.5.10. Failure of the sample to withstand a 1050 pound minimum pull force shall be cause for rejection of previous 4 hours production made on the applicable crimping maching until it is requalified in accordance with paragraph 4.4.3.5.2.

4.4.3.5.2 Alternate testing plan. Each assembly produced shall be subjected to the pull test. The test shall be performed as specified in 4.4.5.10. Failure of any assembly to withstand a 1050 pound minimum pull force shall be cause for rejection of that assembly.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the 'Paragraph Reference/Inspection Method' Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.4.5 Methods of inspection.

4.4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing, or specification requirements, and that all specified manufacaturing processes have been followed.

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4.4.5.2 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3, 4, and 5 of the specification, in accordance with 4.4.2 herein.

4.4.5.3 Moisture.

4.4.5.3.1 First fire composition. The moisture content of the first fire composition shall be determined in accordance with MIL-STD-1234, Method 101.3.1. The special solvent to be used shall be equal parts by volume of dimethylformamide and anhydrous methanol.

4.4.5.3.2 Illuminant composition. The moisture content of illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in MIL-STD-1234, Method 101.2.1 up to paragraph 5.3. A 50 ml. sample of 1:1 dimethylformamide methyl alcohol solution shall be added into a 250 ml. Erlenmeyer flask (fitted with a rubber stopper) containing 10 grams of the pyrotechnic sample. (Exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix the solution and pyrotechnic sample. Allow the solution to set 5 to 15 minutes. A blank sample of 1:1 dimethylformamide methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transferred through the sample porthole into the dry titration beaker containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for at least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary end point is reached. All succeeding operations shall be completed within 3.50 - 0.25 minutes. 25 ml. of the supernatant solution shall be rapidly introduced, accurately pipetted from the supernatant liquid into the beaker through the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 mls) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation:

$$\frac{AB}{5\ (10)} = \text{Percent } H_2O$$

Less _____ Std. Blank

Leaves _____ Percent Moisture in Sample

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Where:
A = mls of Karl Fischer to titrate sample
B = Standardization Factor (mg./ml.)
10 = Constant Factor
5 = Standard Sample Weight

4.4.5.3.3 Alternate method for first fire composition moisture and illuminant composition moisture. The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4, MIL-STD-286. The temperature setting shall be 100 degrees Centigrade and the time setting 30 minutes. The sample size shall be between 0.40 and 0.60 grams.

4.4.5.4 Static functioning test. The test equipment required for this test is delineated in paragraph 2.1.2 under Inspection Equipment Drawings. The temperature of candle shall be 70° ± 5°F at time of test. The average candlepower is to be calculated, using the best 50 seconds of burn time, by dividing the total integral (candlepower-seconds) over this period by 50. The first fire disc assemblies shall be removed prior to testing. The distance from the photometric cell to the flare shell shall be 60.00 ± .25 feet. The axial center of the photometric cell and the axial center of the fireball shall be located 54.0 ± .5 inch from the floor. The temporary back wall listed on dwg. 9201136 shall be removed so that the distance from the center of the candle to the back wall is 14-15 feet. The center line of baffle opening shall be in line with axial center of fireball and axial center of photometric cell. The test shall be conducted in a hearth that has been painted flat black. Depending on ash buildup the tunnel hearth shall be repainted at least once in a three month duration to maintain the black interior or more frequently as determined by the resident Government Quality Assurance Representative. If the average burning time or average candlepower does not meet the requirement of paragraph 4.4.3.2 the illuminant inspection lot represented by the sample shall be rejected.

4.4.5.5 Torque test. The samples supplied in accordance with 4.4.2.7 shall be tested by applying the minimum required torque to the components in the direction of assembly before the thread sealing compound has set. Any perceptible movement of parts will be considered as failure of the assembly to comply with the requirement.

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4.4.5.6 Proving ground. The test shall be performed at a Government Proving Ground. All the cartridges shall be fired from an 81MM, M252 Mortar for which the round is standard. The cartridge shall be assembled with accepted fuzes, fin assemblies, adapters, ignition cartridges, primers, and propellant increments. One half of the sample cartridges shall be fired with one propelling charge increment (charge zone one) and one half of the sample cartridges shall be fired with four propelling charge increments (charge zone four) and in accordance with Table III. However, if meteorological conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to increase the height of burst and insure effective illumination. Observation shall be made for compliance with the requirements specified for security of parts, functioning, functioning time, horizontal range and "effective illumination" (See 6.6). Fuze functioning time shall be reported to at least the nearest 0.10 sec, illuminant burn time to the nearest 0.5 second, and horizontal range on the ground to a point under the position of fuze functioning to the nearest 5 meters. The cartridges shall be conditioned at 70 plus or minus 2.5 degrees Fahrenheit for a minimum of 24 hours prior to testing and fired at this temperature. Any malfunction occurring during the test shall be reported in the official firing record. Meteorological data shall also be recorded in the official firing record. Muzzle velocity shall also be recorded for each round fired and reported in the official firing record.

TABLE III Proving Ground Test

Charge Zone	Fuze Setting (seconds)	Mortar Elevation (degrees)
1	13.5	45
4	33.5	45

4.4.5.7 Security of parts. Parts to be inspected for security shall be manually checked to be tightly and securely attached as described in the applicable drawings. Such parts will not be readily loosened or evidence rattling or other apparent signs of poor assembly when so checked.

4.4.5.8 Strength of heat weld. The sample shall be tested for integrity by applying the required load at the heat weld radially and in a directon away from the axis (center) of the ring.

4.4.5.9 Alternate sampling plan for examination of depth and diameter of rear shear pin holes. The depth and diameter of the rear shear pin holes shall be inspected in the following manner using approved gages.

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4.4.5.9.1 First production shift examination. Beginning with the first production shift, ten (10) consecutive drilled units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every 1/2 hour until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in any of the five (5) units, the machine set up shall be adjusted until ten (10) consecutive units are defect free.

4.4.5.9.2 Second production shift examination. After successfully completing the sampling of paragraph 4.4.5.9.1 for one eight hour shift of production with no defects, the sampling plan may be reduced to five (5) drilled units being inspected every one hour and at the beginning and end of the shift. If a defect is found, the sampling plan of paragraph 4.4.5.9.1 shall be reinstated.

4.4.5.9.3 Third production shift examination. After successfully completing the sampling plan of paragraph 4.4.5.9.2 for one eight hour production shift with no defects, the sampling plan may be reduced to three (3) drilled units being inspected every hour and at the beginning and end of each shift. If a defect is found, the sampling plan of paragraph 4.4.5.9.2 shall be reinstated.

4.4.5.9.4 Fourth production shift examination. After successfully completing the sampling plan of paragraph 4.4.5.9.3 for one eight hour shift with no defects, the sampling plan may be reduced to two (2) drilled units being inspected every two hours. If a defect is found, the sampling plan of 4.4.5.9.3 will be reinstated.

4.4.5.9.5 Tooling change. A change in tooling will require a first piece inspection prior to continuing production. The sampling plan will be the same as that prior to the tooling change.

4.4.5.9.6 Production lapse. A stop in production of over two months will require starting the inspection at the first production shift plan. See 4.4.5.9.1.

4.4.5.10 Pull test for oval sleeve to suspension wire. The test sample shall be assembled on a universal testing machine and the load applied in an axial direction until the required pull force is exerted. The rate of travel of the machine shall be 2 inches per minute maximum.

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5. PACKAGING

5.1 Unit packaging.

5.1.1 Level A. Packaging for cartridges shall be in accordance with the requirements of drawing 9362799.

5.1.2 Level B. Same as level A.

5.1.3 Level C. Not applicable.

5.2 Packing.

5.2.1 Level A. Cartridges shall be packed in accordance with the requirements of drawing 9362798.

5.2.2 Level B. Same as level A.

5.2.3 Level C. Not applicable.

5.3 Marking.

5.3.1 Level A. Packaged cartridges shall be marked in accordance with the requirements of drawing 9362799. Packed cartridges shall be marked in accordance with requirements of drawing 9362798-1.

5.3.2 Level B. Same as level A.

5.3.2 Level C. Not applicable.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for assembly to M853A1 illuminating cartridge for use on 81mm mortar.

6.2 Ordering data. See MIL-A-48078.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.4 Submission of inspection equipment designs for approval. See paragraph 6.2.3 of MIL-A-48078(AR). Submit designs as required to Commander, U.S. Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAT-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.5 Satisfactory functioning. Satisfactory functioning shall be defined as ejection of the parachute and illuminant assembly and proper suspension and ignition of the illuminant such as to assure "effective illumination".

6.6 Effective illumination. Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record.

6.7 Illuminant inspection lot. Illuminant from the mixer will be collected in containers of approximately equal quantity of illuminant which will be numbered and used in sequence. Illuminant candle assemblies will be assigned an inspection lot number for each 85 candles produced. Smaller lot sizes will be accepted when dictated by interruption or shutdown of operations.

6.8 Static burning time. Minimum static burning time as referenced herein (see 3.4) is required, as a time base for the determination of luminosity and compliance does not in itself guarantee meeting the final ballistic minimum burning time requirement. Structural integrity of the candle under set back loading is of prime importance in meeting the ballistic burning time requirement and is attained by proper formulation, blending, consolidation and cure.

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6.9 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following:

Commander
US Army Armament Research, Development and Engineering Center
ATTN: AMSMC-QAT-B (D)/SMCAR-FSA-IM
Picatinny Arsenal, NJ 07806-5000

6.10 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool inspection, statistical process control, substitute sampling plans, ect.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply.

6.11 Subject term (key word) listing.

Cartridge
Functioning
Fuze
Horizontal range
Illuminant
Misfires
Premature
Security
Separation
Static

6.12 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A837)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70544A	Cartridge, 81MM, Illuminating, M853A1 LAP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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